Department of Mathematics and Applied Mathematics
University of Cape Town

Mathematics III **Complex Analysis [3CA]** **2018**

Final Examination

October/November 2018
Time : 2 hours

Marks available: 69
Full marks: $66\frac{2}{3}$

No calculators may be used.

1. Read the following theorem and the outline of its proof:

Theorem. *Let φ be a complex-valued function that is continuous on a simply connected region D. Suppose that γ is a piecewise smooth simple closed path that traces out a positively oriented curve $\mathscr{C}$ in D. For each point z inside $\mathscr{C}$, define*

$$g(z) = \int_\gamma \frac{\varphi(w)}{w - z}\, \mathrm{d}w.$$

Then g is analytic on the interior of $\mathscr{C}$, with

$$g'(z) = \int_\gamma \frac{\varphi(w)}{(w - z)^2}\, \mathrm{d}w.$$

Proof. For each z_0 inside $\mathscr{C}$,

$$g'(z_0) = \lim_{z\to z_0} \int_\gamma \frac{\varphi(w)}{(w - z)(w - z_0)}\, \mathrm{d}w. \tag{1}$$

Put $M = \max_{w\in\mathscr{C}} \left|\varphi(w)\right|$ and $d = \inf_{w\in\mathscr{C}} |w - z_0|$. Note that $d > 0$. If $|z - z_0| < \frac{1}{2}d$, then $|w - z| > \frac{1}{2}d$ for all $w \in \mathscr{C}$. For such z,

$$\left|\int_\gamma \frac{\varphi(w)}{(w - z)(w - z_0)}\, \mathrm{d}w - \int_\gamma \frac{\varphi(w)}{(w - z_0)^2}\, \mathrm{d}w\right| \leqslant \frac{2M}{d^3}\, |z - z_0|\, |\mathscr{C}|. \tag{2}$$

Thus $\displaystyle\int_\gamma \frac{\varphi(w)}{(w - z)(w - z_0)}\, \mathrm{d}w \to \int_\gamma \frac{\varphi(w)}{(w - z_0)^2}\, \mathrm{d}w$ as $z \to z_0$, as required. $\square$

Now answer the following questions:

(a) Show how to derive equation (1).

(b) How do we know that $M = \max_{w\in\mathscr{C}} \left|\varphi(w)\right|$ exists?

(c) How do we know that $d > 0$?

(d) Prove that if $|z - z_0| < \frac{1}{2}d$, then $|w - z| > \frac{1}{2}d$ for all $w \in \mathscr{C}$.

(e) Show how to derive inequality (2). [3, 2, 2, 2, 6]

2. Let $f(z) = \left|z^2 - 4\right|^2$.
   (a) Is f differentiable at $z = 2$? (A "yes" or "no" answer will not suffice.)
   (b) Are the Cauchy–Riemann equations satisfied at $z = 2$? (Again, "yes" or "no" is not enough.)
   [5]

3. Find the radius of convergence of each of the following power series.

$$\text{(a)}\quad \sum_{k=0}^{\infty} \left(3 + (-2)^k\right)^k z^k \qquad\qquad \text{(b)}\quad \sum_{k=1}^{\infty} \frac{(2k)^k}{k!}(z-3)^k$$

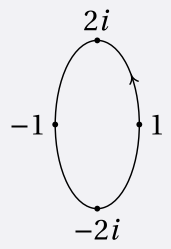
[3, 6]

4. Does there exist a function f that is analytic on the open disc $D_1 = \{z \in \mathbb{C} : |z| < 1\}$ and that satisfies

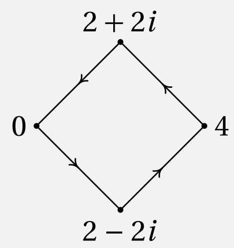
$$f\left(\frac{1}{n}\right) = \frac{(-1)^n}{n}$$

for all integers $n \geqslant 2$? (A "yes" or "no" answer won't earn you any marks: you need to give a proper reason.)
[3]

5. Find (a) $\displaystyle\int_\gamma \frac{e^{\pi z}}{z(z^2+1)}\,dz$, where γ traces out the curve



(b) $\displaystyle\int_\gamma \frac{1+\cos z}{(z-\pi)^3}\,dz$, where γ traces out the curve



[6, 4]

6. Use complex variable methods to evaluate

$$\text{(a)}\quad \int_0^{2\pi} \frac{1}{13 + 5\sin\theta}\,d\theta \qquad\qquad \text{(b)}\quad \int_0^{\infty} \frac{x^2+1}{x^4+1}\,dx$$

[7, 7]

7. Show that all the zeros of $z^7 - 5z^3 + 12$ lie in the annulus $\{z \in \mathbb{C} : 1 < |z| < 2\}$. [7]

8. Find the Laurent series expansion of the form $\displaystyle\sum_{k=-\infty}^{\infty} a_k z^k$ for the function $f(z) = \dfrac{5z-7}{z^2-2z-3}$ that converges when $z = 2$. What is the domain of convergence of this series? [6]